

Alright, let's rearrange the equation to solve for the initial velocity, U , in terms of the horizontal distance x , terminal velocity V , gravity g , and time t .

Here's the original equation:

$$x = \frac{V^2}{g} \ln\left(\frac{V^2 + gUt}{V^2}\right)$$

Here are the steps to isolate U :

1. Multiply both sides by $\frac{g}{V^2}$:

$$\frac{gx}{V^2} = \ln\left(\frac{V^2 + gUt}{V^2}\right)$$

2. Exponentiate both sides using the base e (the natural logarithm's inverse):

$$e^{\frac{gx}{V^2}} = \frac{V^2 + gUt}{V^2}$$

3. Multiply both sides by V^2 :

$$V^2 e^{\frac{gx}{V^2}} = V^2 + gUt$$

4. Subtract V^2 from both sides:

$$V^2 e^{\frac{gx}{V^2}} - V^2 = gUt$$

5. Factor out V^2 on the left side:

$$V^2 (e^{\frac{gx}{V^2}} - 1) = gUt$$

6. Finally, divide both sides by gt to solve for U :

$$U = \frac{V^2 (e^{\frac{gx}{V^2}} - 1)}{gt}$$

So, the initial velocity U expressed in terms of x , V , g , and t is:

$$U = \frac{V^2}{gt} (e^{\frac{gx}{V^2}} - 1)$$

Let me know if you'd like to use this equation to solve for a specific value of U given certain values for the other variables!